Mohammadreza Narimani

Artificial Intelligence | Computer Vision | Digital Agriculture | Precision Sensing Systems

PhD Candidate in Biological Systems Engineering

University of California, Davis

Biological and Agricultural Engineering Department

California, US

Website: mohammadrezanarimaniucdavis.github.io
GitHub: MohammadrezaNarimaniUCDavis
Scholar: Google Scholar Profile
Email: mnarimani@ucdavis.edu
Personal: narimanimohammadreza@gmail.com

Research Interests

Remote Sensing of Agriculture and Environment • Earth Observation • Digital Agriculture • Machine Learning • Computer Vision • Precision Agriculture • Hyperspectral Imaging • LiDAR • Satellite Data Analysis • IoT Systems • Plant Phenotyping • 3D Radiative Transfer Modeling • Wildfire Detection

Education

Ph.D. Candidate of Biological Systems Engineering — 2022–Present

University of California, Davis, US — **GPA: 4.0/4.0**
Thesis: AI-Driven Remote Sensing and Spectral Modeling for Advanced Monitoring of Tomato Crops
Advisors: Prof. Alireza Pourreza, Prof. Ali Moghimi

M.Sc. of Biosystem Mechanical Engineering — 2019–2021

University of Tehran, Iran — **GPA: 4.0/4.0 (First Rank)**
Thesis: Designing and Manufacturing an Experimental Smart Greenhouse for High-Throughput Stress Phenotyping and Plant Disease Detection by Implementing Internet of Things System and Machine Learning

B.Sc. of Biosystem Mechanical Engineering — 2015–2019

University of Tehran, Iran — **GPA: 4.0/4.0 (First Rank)**
Thesis: Design and construction of electric turning machine for urban agriculture

Research Experience

Research Assistant — 2022–Present

Digital Agriculture Laboratory, University of California, Davis

- AI-Driven Remote Sensing and Spectral Modeling for Advanced Monitoring of Tomato Crops
- 3D Radiative Transfer Modeling for Specialty Crops

- Wildfire Detection and Monitoring Using Satellite Imagery and Deep Learning
- Image semantic segmentation using FCN_AlexNet, SegNet, and U-Net for Agricultural Vision Competition

Research Assistant 2016–2022

University of Tehran, Iran

- High-Throughput Wheat Phenotyping for Agricultural Research Institute of Iran, Wheat Species Bank
- Detecting fungal disease of lettuce with thermal camera and CNNs
- Image processing algorithms for measuring friction coefficient and angle of repose in rice grain
- IoT device development using Arduino, Raspberry Pi, Ethernet, and ESP8266
- Designing novel fogponic and centrifugal irrigation systems for aeroponic greenhouses
- Deep learning algorithms for detecting powdery tomato disease
- Designing strawberry harvester robot

Internship 2017–2022

Agricultural Engineering Research Institute of Iran (AERI)

Technical Skills

Remote Sensing Tools: Sentinel-2, Landsat-9, EMIT, ECOSTRESS, DJI Phantom 4 Multispectral, Aerial Pika-L Hyperspectral Sensor, Aerial Phoenix LiDAR Systems

Programming Languages: Python, R, MATLAB, C++

Amazon Web Services: Lambda, S3, App Runner, API Gateway, EC2, Elastic Container Registry, SageMaker, Bedrock, IAM, CloudFormation, CloudWatch

Techniques: Computer Vision, Image Processing, Web App Development, Internet of Things, Robotics

Prominent Libraries: Keras, TensorFlow, PyTorch, scikit-learn, geemap, rasterio, gdal, pandas, NumPy, OpenCV, PlantCV, scikit-image

GIS: ArcGIS, QGIS, Google Earth Engine, ENVI, Pix4D, Resonon, Spatial Explorer, Microstation, Terrasolid

Computer-Aided Design and Engineering: CATIA, SolidWorks, AutoCAD, Abaqus, Ansys, Keyshot

Artificial Intelligence

Deep Learning: Instance segmentation (Mask R-CNN, YOLO, SAM); semantic segmentation of images (U-Net, FCN, SegNet, DeepLab); image classification and object detection; frameworks: PyTorch, TensorFlow, Keras

Machine Learning: Regression (linear, polynomial, ridge, LASSO, Gaussian process); classification (Random Forest, SVM, XGBoost, logistic regression, k-NN, decision trees, gradient boosting); model selection and validation; scikit-learn

Large Language Models (LLMs): GPT-4, Claude (Anthropic), Llama (Meta), Amazon Bedrock, Kiro; prompt engineering, RAG (retrieval-augmented generation), API integration and deployment

Journal Publications

1. **Narimani, M.R.**, Pourreza, A., Farajpoor, P. (2026). Mapping Tomato Cropping Systems in California Using AlphaEarth Geospatial Embeddings and Deep Learning Analysis. *arXiv preprint.* DOI Link
2. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Farajpoor, P. (2026). Sentinel-2 for Crop Yield Estimation: A Systematic Review. *arXiv preprint.* DOI Link
3. Farajpoor, P., Pourreza, A., **Narimani, M.R.**, El-Kereamy, A., Fidelibus, M. (2025). Multi-Trait Spectral Modeling for Estimating Grapevine Leaf Traits and Nutrients. *Plant Phenomics.* DOI Link
4. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Farajpoor, P., Jafarbiglu, H., Mesgaran, B. (2025). Early Detection of Branched Broomrape (Phelipanche Ramosa) Infestation in Tomato Crops by Using Leaf Spectral Analysis and Machine Learning. *IFAC-PapersOnLine.* DOI Link
5. Alimardani, R., Adedeji, A.A., **Narimani, M.R.** (2025). A Review of IoT Applications in Food Processing and Related Fields. *Journal of Food Processing and Preservation.* DOI Link
6. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Farajpoor, P., Jafarbiglu, H., Mesgaran, B. (2025). Branched broomrape detection in tomato farms using satellite imagery and time-series analysis. *SPIE Defense + Commercial Sensing.* DOI Link
7. Farajpoor, P., Pourreza, A., **Narimani, M.R.**, El-Kereamy, A., Fidelibus, M. (2024). Leaf spectral reflectance prediction using multihead attention neural networks. *SPIE Defense + Commercial Sensing.* DOI Link
8. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Mesgaran, B., Farajpoor, P., Jafarbiglu, H. (2024). Drone-based multispectral imaging and deep learning for timely detection of branched broomrape in tomato farms. *SPIE Defense + Commercial Sensing.* DOI Link
9. Akpenpuun, T.D., Ogunlowo, Q.O., Na, W.H., Dutta, P., Rabiu, A., Adesanya, M.A., **Narimani, M.R.**, et al. (2024). Dynamic neural network modeling of thermal environments of two adjacent single-span greenhouses. *Journal of Agricultural Engineering.* DOI Link
10. **Narimani, M.R.**, Hajiahmad, A., Moghimi, A., Alimardani, R., Rafiee, S., Mirzabe, A.H. (2021). Developing an aeroponic smart experimental greenhouse for controlling irrigation and plant disease detection using deep learning and IoT. *ASABE Journal.* Publication Link
11. Kianmehr, M., Elyasi, M., **Narimani, M.R.** Design and construction of electric turning machine for urban agriculture. *Journal of Tarbiyat Modares University.* Publication Link

12. Moghadam, S., Alimardani, R., **Narimani, M.R.**, Moghimi, A. Internet of Things and Artificial Intelligence for smart agriculture: a systematic mapping review of sensors, platforms, connectivity, and machine learning applications (2014-2024). *AgriRxiv.* DOI Link

Conference Presentations

1. **Narimani, M.R.**, et al. (2025). Early Detection of Branched Broomrape Infestation in Tomato Crops. *8th IFAC Conference on Sensing, Control and Automation Technologies for Agriculture.* Conference Link
2. **Narimani, M.R.**, et al. (2024). Drone-based multispectral imaging and deep learning for timely detection of branched broomrape. *SPIE Defense + Commercial Sensing.* Presentation Link
3. **Narimani, M.R.**, et al. (2024). Early Detection of Branched Broomrape in Tomato by Hyperspectral Sensing. *ASABE 2024.*
4. Jafarbiglu, H., Pourreza, A., **Narimani, M.R.**, Sanden, B., Marino, G., Culumber, M., Mehata, M., Ferguson, L. (2024). Determining the best irrigation strategy for pistachio orchards with saline water and soil. *SPIE Defense + Commercial Sensing.* Presentation Link
5. **Narimani, M.R.**, et al. (2024). Remote Sensing of Branched Broomrape in Tomato. *65th Weed Day, UC Davis.* Event Link
6. **Narimani, M.R.**, et al. (2023). Wildfire Detection and Monitoring Using Satellite Imagery and Deep Learning. *ASABE 2023.* Schedule Link
7. Jafarbiglu, H., Pourreza, A., **Narimani, M.R.**, Williams, D., Ferguson, L. (2023). Determining the best irrigation practices for pistachio orchards with saline water and soil. *ASABE 2023.* Schedule Link
8. **Narimani, M.R.**, Hajiahmad, A., Moghimi, A., Alimardani, R., Rafiee, S., Mirzabe, A.H. (2021). Developing an aeroponic smart experimental greenhouse for controlling irrigation and plant disease detection using deep learning and IoT. *ASABE 2021.*

Conference Posters

1. Farajpoor, P., Pourreza, A., **Narimani, M.R.** (2025). Domain Adaptation Approaches to Improve Leaf Trait Estimation by Reducing Error Propagation in Hybrid Modeling. *AGU 2025.*
2. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Farajpoor, P., Jafarbiglu, H., Mesgaran, B. (2025). Branched broomrape detection in tomato farms using satellite imagery and time-series analysis. *SPIE Defense + Commercial Sensing.* Poster Link
3. Jafarbiglu, H., Pourreza, A., **Narimani, M.R.**, Bailey, J., Taylor, G. (2024). Seam Carving for Tree Segmentation in Dense Tree Farms. *ASABE 2024.*
4. **Narimani, M.R.**, Hajiahmad, A., Moghimi, A., Alimardani, R., Rafiee, S., Mirzabe, A.H. (2021). Developing an aeroponic smart experimental greenhouse for controlling irrigation and plant disease detection using deep learning and IoT. *ASABE 2021.*

Teaching Experience

Teaching Assistant, University of California, Davis 2022–Present

- Applied Statistics in Agricultural Sciences Lab Materials
- Environmental Analysis Using GIS Lab Materials
- Environmental Monitoring
- Introduction to Unmanned Aerial Systems for Agriculture & Environmental Science

- Communications and Computing Technology (IoT in Ag) Lab Materials

Teaching Assistant, University of Tehran, Iran 2016–2022

- Programming with Python and MATLAB
- Computer Vision and Artificial Intelligence
- Industrial drawing with CATIA, SolidWorks and AutoCAD

Teaching Physics, Bonyad Amoozeshi Ghalamchi 2014–2016

Awards and Honors

- First Rank, Excellence in Specialty Crops Award, Farm Robotics Challenge (Selected from 96 global teams) 2026
- Ernest E. Hill Memorial Graduate Fellowship, UC Davis 2026–2027
- Walter Rosenberg Research Fellowship (GBSE Fellowship), UC Davis 2026
- Future Undergraduate Science Educators (FUSE) Program Scholarship 2025–Present
- Best Presentation Award, ASABE Annual International Meeting 2024
- Agricultural Genome to Phenome Initiative Scholarship 2024
- Bill And Jane Fischer Vegetation Management Scholarship 2024
- Member of SPIE Defense + Commercial Sensing 2024–Present
- Peter J. Shields and Henry A. Jastro Research Award 2023
- Dean's Distinguished Graduate Fellowship 2022
- Member of ASABE (American Society of Agricultural and Biological Engineers) 2021–Present
- First rank prize, Iran IoT and Computer Vision Competition 2020
- Member of Iran Elite Foundation (Bonyad Melli Nokhbegan) 2017–2022

Leadership and Service

Leadership Positions

- Graduate Student Leader, UC Davis Team, Farm Robotic Challenge, UCANR Innovate & AIFS 2025–Present
- Vice President, Graduate Student Association, BAE Department, UC Davis 2024–Present
- Representative, Graduate Student Association, BAE Department, UC Davis 2023–2024
- Election Committee Member, Graduate Student Association, BAE Department 2025

Mentorship

- Mentor, E-SEARCH Program, UC Davis (4 undergraduate students) 2024–Present
- Mentor, AgTech Workshop (Drone & IoT), AI Institute for Next Generation Food Systems 2023–Present
- Mentor, Young Scholar Program, UC Davis (high school students) 2023

Mentees' Research Posters

- Sun, V.J., **Narimani, M.R.** Deep Learning-Based Snow Monitoring in California Using Sentinel-2 Satellite Data. ESEARCH Summer 2025.

- Kumar, S., **Narimani, M.R.**, et al. Enhancing Orchard Management with Deep Learning: Tree Segmentation Using Geospatial SAM2 Model. ESEARCH Fall 2024.
- Richmond, N., **Narimani, M.R.**, et al. Global Vegetation and Climate Insights Portal (GVCIP). ESEARCH Summer 2024.
- Tran, Q., **Narimani, M.R.**, et al. Enhancing Wildfire Monitoring Through Remote Sensing. ESEARCH Spring 2024.

Editorial and Peer Review Activities

Reviewer for the following journals: 2024–Present

- Scientific Reports (Springer Nature)
- Journal of Agriculture and Food Research (Elsevier)
- Journal of Food Science (Wiley)
- Smart Agricultural Technology (Elsevier)
- Computers and Electronics in Agriculture (Elsevier)
- Information Processing in Agriculture (Elsevier)
- Remote Sensing Applications: Society and Environment (Elsevier)
- NJAS: Impact in Agricultural and Life Sciences (Taylor & Francis)

Media Coverage

Parasitic Weeds Threaten Tomato Plants on California Farms - Featured in:

- UC Davis Newsletter Read Article
- UC Davis College of Agricultural and Environmental Sciences Read Article
- Daily Democrat Newspaper Read Article
- Ag Alert Magazine Read Article
- Capital Press Newsletter Read Article
- Organic Growers Newsletter Read Article
- UC Davis Plant Sciences Newsletter Read Article
- Tomato News Read Article
- California Fruit and Vegetable Read Article

First Rank, Excellence in Specialty Crops Award, Farm Robotics Challenge 2026 - Featured in:

- UCANR News Release (Awards at USA Plug and Play Summit) Read Article
- Farm Robotics Challenge Official 2026 Results View Results
- Project BloomSense Official Video Entry Watch Video

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